

Ganesh T

Senior AI/ML Engineer

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PROFESSIONAL SUMMARY

Senior **AI/ML Engineer** with 8+ years of experience in designing and deploying **machine learning** and **generative AI solutions** for **healthcare, finance, and e-commerce**. Strong hands-on expertise in **LLMs, RAG pipelines, Agentic AI, vector databases, and automation**. Skilled in **Python, cloud platforms (AWS & Azure), MLOps, scalable data pipelines, and model deployment**. Proven track record of improving system performance, delivering actionable business insights, and reducing processing time. Experienced in mentoring junior engineers, collaborating with teams, and turning AI solutions into practical business value. Passionate about using AI to solve complex problems and drive innovation.

TECHNICAL SKILLS

Category	Skills / Tools
ML & AI Frameworks	PyTorch, Hugging Face Transformers, TensorFlow, Keras, Scikit-Learn, XGBoost, LightGBM, spaCy, Sentence Transformers, LangChain, LangSmith, Linear/Logistic Regression, Decision Trees, Random Forest, Classification, Tabular Data Processing
Generative AI & NLP	LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, LoRA, QLoRA, PEFT, OpenAI GPT-4/3.5, Anthropic Claude, AutoGen, Amazon Bedrock, Mistral, Vector Databases (Amazon OpenSearch, Pinecone), Agentic AI, LLM integration
Data Engineering & ETL	Python, Apache Spark, Kafka, Airflow, AWS Glue, Azure Data Factory, Delta Lake, PostgreSQL, MySQL, Elasticsearch, PySpark, Pandas, NumPy, AWS S3
MLOps & LLMops	AWS SageMaker, Azure ML, Databricks, MLflow, CI/CD (GitHub Actions, Jenkins, Terraform), Prometheus, Grafana
Development & Deployment	Python, FastAPI, Flask, REST APIs, Streamlit, Jupyter, Docker, Kubernetes (AWS EKS / Azure AKS), Terraform, Git/GitHub
Cloud Platforms	AWS: SageMaker, S3, RDS, DynamoDB, Glue, Athena, Redshift, Lambda, EC2; Azure: Azure ML, Blob Storage, Data Factory, Cognitive Services, AKS, Synapse Analytics, Azure DevOps
Collaboration Tools	Jira, Confluence, Agile/Scrum, Trello, Lucidchart, Draw.io

WORK EXPERIENCE

Senior AI Engineer | Geico
Nov 2023 – Present | Dallas, TX

Project Description:

Architected and deployed enterprise-grade Agentic AI solutions for insurance operations at GEICO. Designed production **RAG architecture** with multi-agent orchestration using **LangChain, LangGraph, OpenSearch, and Pinecone**. Built robust LLM evaluation frameworks with **LangSmith and custom guardrail pipelines** to ensure safe, hallucination-free AI deployments across claims triage, policy Q&A, and underwriting.

Roles & Responsibilities:

- Led **GenAI Engineering** efforts to design and deploy end-to-end **LLM-powered** intelligent automation for insurance claims and policy operations, improving retrieval accuracy by 25% and reducing query resolution time by 40%.

- Designed production-grade **RAG Architecture** with hybrid retrieval (**semantic + keyword**), optimized chunking strategies, and reranking pipelines using **Amazon OpenSearch and Pinecone**, reducing average document retrieval latency by 35% and enabling context-aware policy Q&A across 2M+ insurance documents.
- Built and managed Agent Orchestration frameworks using **LangChain and LangGraph**, enabling autonomous multi-step claims processing across 5+ enterprise insurance systems, reducing manual claims handling effort by 45% and cutting end-to-end processing time by 30%.
- Implemented comprehensive LLM Evaluation Frameworks using **LangSmith and RAGAS-aligned** custom metrics (faithfulness, context recall, answer relevance), reducing hallucination rate by 60% and achieving 92% answer relevance score in production.
- Developed advanced AI agents with **autonomous decision-making for claims triage, enabling multi-agent orchestration** with guardrails automating 70% of routine claims decisions and reducing human review queue by 3x.
- Fine-tuned LLMs using **PEFT (LoRA/QLoRA)** on 500K+ insurance-domain records, reducing hallucinations in claims summarization by 55% and improving domain-specific prediction accuracy by 18%.
- Built **Python + FastAPI microservices exposing RAG and GenAI** capabilities via a unified API Gateway, achieving <200ms p95 latency and supporting 300+ concurrent users across underwriting and claims teams.
- Automated ingestion pipelines from 4+ sources (**SharePoint, ServiceNow, AWS S3, policy documents**) using **Python + Jenkins**, processing 50K+ documents daily and cutting data onboarding time by 65%.
- Established LLMOps observability using **Prometheus and Grafana**, achieving 99.5% system uptime and reducing mean time to resolution (MTTR) for production incidents by 50%.
- Mentored junior engineers and collaborated with business stakeholders and product managers to deliver enterprise-grade GenAI solutions, translating AI agent insights into actionable strategies and measurable business impact

Key Tools & Technologies: Python, FastAPI, LangChain, Sentence Transformers, OpenAI (Azure), Anthropic (Bedrock), Mistral, Hugging Face, LoRA, QLoRA, PEFT, Amazon OpenSearch, AWS S3, Jenkins, Terraform, Azure AKS, Prometheus, Grafana, LangSmith, MLflow, Jira, Confluence, SharePoint, ServiceNow, Design of Experiments (DOE)

Machine Learning Engineer | Atos
Jul 2021 – Aug 2023 | Fremont, CA

Project Description:

Designed and delivered an end-to-end Azure-native ML solution to automate **support ticket triage** and analyze patient/provider feedback sentiment. Operationalized scalable **MLOps** workflows on large-scale healthcare data, reducing resolution times, improving customer satisfaction, and ensuring reliable, reproducible ML processes.

Roles and Responsibilities:

- Designed and developed scalable ETL pipelines using **Azure Data Factory** and **Databricks (PySpark)** to process **10M+** healthcare support tickets, generating ML-ready datasets for predictive models.
- Developed and deployed an automated **ticket prioritization system** using **regression** and **classification models**, leveraging **Azure Cognitive Services** and custom **NLP classifiers**, incorporating temporal metadata features and drift monitoring, reducing critical ticket resolution time by **35%**.
- Enhanced sentiment analysis accuracy using **Azure Text Analytics** combined with domain-specific NLP models.
- Developed **ML models** and **anomaly detection** workflows within the support analytics platform to uncover irregular patterns in patient-provider interactions and **claims data**, reducing **fraudulent activities** by **28%** and strengthening fraud prevention and payment integrity initiatives.
- Implemented **MLOps lifecycle** with Azure ML, DevOps, Docker and AKS, enabling real-time, low-latency inference.
- Built serverless APIs with **Azure Functions** and managed them via **Azure API Management** for smooth integration.
- Ensured HIPAA-compliant data governance with secure **Azure Blob Storage**, RBAC, and automated compliance checks, reducing audit prep time by **20%**.
- Created **Power BI dashboards** to visualize sentiment trends, ticket resolution KPIs, and SLA compliance, and established reusable ML/data engineering frameworks to accelerate future NLP projects.

Key Tools & Technologies: Python, SQL, PySpark, scikit-learn, PyTorch, spaCy, NLTK, Azure Machine Learning, Azure Cognitive Services (Text Analytics, Sentiment Analysis, NER), Azure Databricks, Azure Data Factory, Azure Blob Storage, Azure Synapse Analytics, Docker, Azure Kubernetes Service (AKS), Azure Functions, Azure API Management, GitHub, Azure DevOps, Terraform, Azure Monitor, Application Insights, Power BI.

Data Scientist | Fidelity Investments
Oct 2019 – Jun 2021 | Merrimack, NH

Project Description:

Worked on building a credit risk analytics platform to better predict credit card defaults and support smarter lending decisions. Focused on handling large-scale, imbalanced financial data and delivering reliable risk insights to business teams.

Roles and Responsibilities:

- Built credit risk classification models using **XGBoost** to identify high-risk customers, improving default prediction accuracy and reducing missed risk cases.
- Addressed class imbalance using **SMOTE**, which improved recall for high-risk customers and made the model more reliable in real-world scenarios.
- Evaluated and fine-tuned models using metrics like **ROC-AUC**, precision-recall, and threshold optimization to align with business risk objectives.
- Developed scalable data pipelines using **Python and PySpark** to process large volumes of transaction data, reducing processing time from ~24 hours to a few hours.
- Implemented real-time data ingestion with **Kafka and AWS Kinesis** to enable near real-time risk scoring and faster decision-making.
- Built dashboards and automated workflows using **Tableau, Airflow, and Docker**, improving visibility for stakeholders and reducing model deployment time significantly.

Key Tools & Technologies: Python, PySpark, Apache Kafka, AWS Kinesis, AWS S3, AWS Glue, Amazon Redshift, Docker, PostgreSQL, Apache Airflow, Cron, Elasticsearch, Tableau, AWS QuickSight, CloudWatch, Kibana, Git, REST APIs, CI/CD

Python Developer | Reliance Retail
May 2017 – Sep 2019 | Mumbai, India

Project Description:

Developed Python automation scripts, APIs, and visualization tools to streamline data processing across Reliance's retail, energy, and telecom business units. Laid the data engineering foundation that enabled subsequent transition into ML engineering roles.

Roles and Responsibilities:

- Built 10+ scalable **RESTful APIs with Flask** to deliver KPIs in real time reducing reporting times from 3 days to under 1 hour and saving an estimated 200+ manual hours per month across business units.
- Created **Python automation and ETL pipelines** using **Pandas and SQL Alchemy** to process 5M+ customer interaction records weekly, reducing manual data processing workload by 70% and eliminating 3 full-time manual reporting tasks.
- Developed 5+ interactive **Tableau** dashboards for self-service analytics and optimized 50+ **SQL queries** reducing average query execution time by 65% and enabling non-technical teams to independently access business insights.
- Implemented unit and integration test suites with **PyTest** achieving 85%+ code coverage and created 12 reusable Python libraries reducing bug rate by 40% and cutting new feature development time by 25%.
- Designed and maintained 15+ database schemas in **PostgreSQL and MySQL**, wrote optimized stored procedures supporting BI reporting for 3 divisions, reducing report generation time by 55% and supporting \$2M+ in annual data-driven decisions.

- Collaborated with cross-functional teams across retail, telecom, and energy units to gather requirements, translate business needs into technical solutions, and deliver Python-based tools that improved operational efficiency by 30%.

Key Tools & Technologies: Python, Flask, Django, Pandas, SQLAlchemy, PostgreSQL, MySQL, Tableau, Matplotlib, Seaborn, PyTest, unittest, Linux scripting, Git, REST APIs, CI/CD, Jenkins

CERTIFICATIONS & EDUCATION

- AWS ML – Specialty
- AWS Cloud Quest – Generative AI Architect
- Microsoft Azure AI associate
- Databases and SQL for Data Science - IBM
- Generative AI fundamentals - Databricks
- Generative AI with Large Language Models - DeepLearning.AI

EDUCATION

- Sathyabama Institute of Science and Technology | Bachelor of Technology | Computer Science and Engineering
- University of North Texas | Masters of Science | Data Engineering